
agent-location-service (:4004, Go)

Request flow, file by file

150 questions with answers

12 sections

Event-driven microservices · Go · Kafka · Redis

7 — agent-location-service (:4004, Go)

Job: where every on-shift agent is, right now. The highest-frequency write in the system.

```
cmd/server/main.go
├ config.Load()                internal/config/config.go
├ store.New(redisURL, agentTTL) internal/store/redis.go
├ api.Routes(handlers, secret) internal/api/router.go
├ go runReaper(...)           ▶ background goroutine
└ server.ListenAndServe()

— POST /agents/location {lat,lng} —————
▼
internal/api/router.go
└ agentOnly( ... )           internal/httpx/middleware.go
  ▶ trace id → context
  ▶ verify JWT, pin HS256
  ▶ require role AGENT
▼
internal/api/handlers.go UpdateLocation()
├ decode body; validateCoordinates()
│   ▶ lat/lng are *float64 POINTERS, so a missing field is
│   told apart from a real 0 (the equator / Greenwich)
├ ensureCapabilities()        ← the Phase 13 fix
│   └ store.HasCapabilities()
│       HEXISTS agent:<id>:caps updatedAt
│       ▶ NOT plain EXISTS – a hash created by the
│       availability toggle is a stub, not a mirror
│   └ internal/authclient/client.go → auth-service /auth/me
├ store.SaveCapabilities()    full mirror, writes updatedAt
└ store.SavePartialProfile()  auth down: name/phone only,
                              NO updatedAt → stays retryable

├ store.UpsertLocation()      one Redis pipeline:
│   GEOADD agents:live <lng> <lat> <agentId>
│   SET    agent:<id>:alive 1 EX 120
│   (!) GEOADD takes LONGITUDE FIRST. Reversing them is silent.
└ internal/httpx/respond.go   JSON 200 { expiresInSeconds: 120 }
```

The other endpoints, and the reaper

```
— the other endpoints —————
POST /agents/availability → ensureCapabilities() FIRST, then
                           HSET caps available
POST /agents/offline      → ZREM agents:live + DEL alive
GET  /agents/nearby       → GEOSEARCH (what the engine would see)

— background goroutine —————
runReaper → store.Reap()
  ▶ a sorted-set MEMBER cannot expire, only a whole key.
    So: the alive key expires (checked on every read), and the
    reaper removes orphaned geo members so the set cannot grow
    without bound.
```

The four Redis keys

agents:live	GEO set	every position, one member per agent
agent:<id>:caps	HASH	capabilities, rating, contact, updatedAt
agent:<id>:load	counter	pending pickups (INCR/DECR, atomic)
agent:<id>:alive	string+TTL	heartbeat, 120s, refreshed every 20s

`assignment-engine` **reads these keys directly** — the one deliberate boundary exception in the system, because an HTTP hop on the hot path would cost more than it is worth. The key layout is a contract between the two services.
